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Hierarchical 3D Reduced Graphene Porous-Carbon-Based PCMs for Superior Thermal Energy Storage Performance

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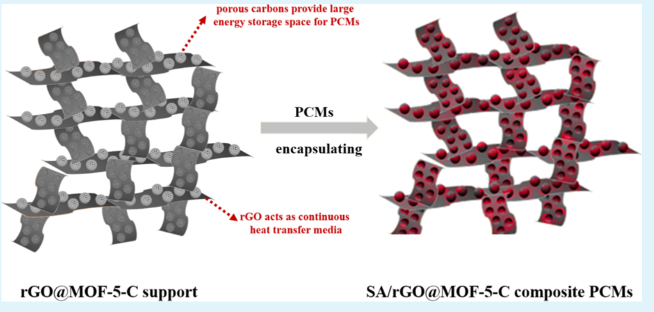
S Supporting Information

ABSTRACT: Phase change enthalpy and thermal conductivity are the two essential parameters for practical applications of shape-stabilized phase change materials (ss-PCMs). Herein, hierarchical three-dimensional (3D) reduced graphene porous carbon support PCMs have been successfully synthesized by carbonizing a graphene oxide@metal−organic framework (GO@MOF) template, which simultaneously realizes large phase change enthalpy and high thermal conductivity. During the carbonization process, MOFs were converted into hierarchical porous carbons, whereas GO was reduced to high-thermal-performance reduced graphene (rGO). Thus, a hierarchical 3D porous carbon structure with high porosity and large specific surface area was obtained, which provided a suitable condition for encapsulating PCMs. Furthermore, the pores of carbon stabilized the PCMs by capillary force and surface tension. The interaction between the PCM molecule and rGO significantly decreased the interfacial thermal resistance and made the composites reveal high thermal conductivity. Furthermore, the 3D network structure promoted the stretching and crystallization characteristics of the stearic acid molecule in the confined pore space, which enhanced the heat release efficiency. Compared with the rGO/MOF-5-C support, the hierarchical 3D structure of rGO@MOF-5-C revealed a thermal conductivity of 0.60 ± 0.02 W m^{-1} K^{-1}, which was 27.7% improvement, with large phase change latent heat of 168.7 J g^{-1}, which increased by 18.5%. Additionally, the obtained ss-PCMs showed transient thermal response and good durability, indicating its promising potential in thermal energy storage application.



KEYWORDS: *phase change materials, reduced graphene oxide, 3D network structure, phase change enthalpy, thermal conductivity, metal−organic frameworks*

INTRODUCTION

Organic phase change materials (PCMs) have attracted extensive interest among energy storage materials owing to their excellent capacity of absorbing and releasing latent heat during a phase change process.[1−5] Particularly, the small volume change, chemical stability, and constant phase transition process make the organic PCMs a promising material for waste heat recovery and electrical and solar energy applications.[6−12] For example, fatty acids and paraffin have been widely investigated for solar energy application due to their high phase change enthalpy, almost negligible supercooling, and excellent thermal stability.[13] However, some inherent defects, such as leakage and low heat conduction in pure organic PCMs, limit their large-scale applications.[14] Extensive efforts have been made to impregnate organic PCMs into solid porous supporting materials to overcome the challenges. For instance, porous materials like metal−organic framework,[15] diatomite,[16−18] perlite,[19−22] expanded graphite,[23−26] and active carbon[27] combined with low cost, simple production step, and excellent thermal stability make the materials a promising solid support for PCMs. Highly porous materials with a large specific surface area can stabilize the PCMs through surface tension and capillary forces. However, highly porous materials usually have amorphous structures, which severely limit thethermal conductive efficiency.[28] Moreover, PCM molecules restricted in confined pores present a lower phase change enthalpy.[29−31] It is still a challenge to develop advanced PCMs with integrated enhanced heat conduction and energy storage capacity for real applications.

Recently, graphene and graphene oxide (GO) have been applied to enhance the thermal energy storage performance of composite PCMs. For example, Zamiri et al. added 4%

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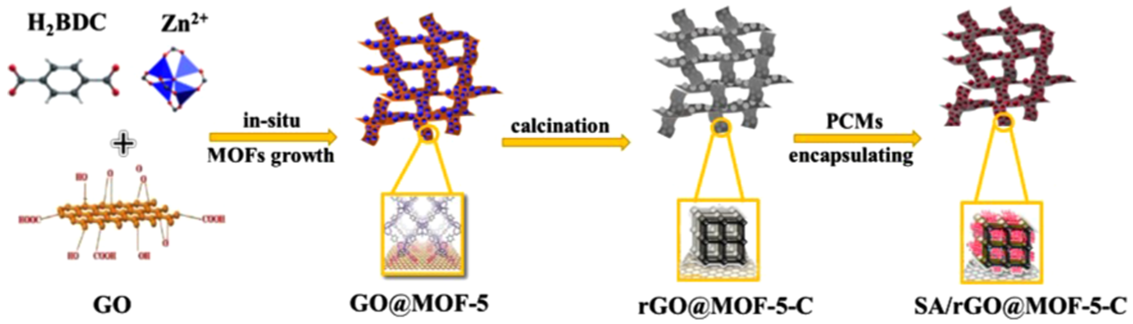
Scheme 1. Schematic Illustration of ss-PCM Composite Synthesis



graphene into the PCM matrix and the thermal transport performance of the composites was increased about 1.5-fold.[32] Meanwhile, Mehrali et al. developed a commercial graphene and stabilized 91.9 wt % PCMs with about 10 times heat conduction improvement than that of the pure PCM.[33] In addition, three-dimensional (3D) graphene foam and graphene aerogels were further developed to increase the thermal performance of ss-PCMs.[34,35] The supports offer a large number of continuously interconnected open pores and absorb more PCMs with a significant improvement in the thermal conductivity. Recently, cost-effective and facilely fabricated graphene oxide (GO) has been extensively exploited.[36,37] Li[38] and Yang[39] developed GO to stabilize poly(ethylene glycol) and exhibited large energy storage density. However, the low heat conduction capability of GO limits its heat transfer efficiency. Currently, rGO derived from physical or chemical reduction of GO is an ideal thermal conductive material, which removed most of the oxygen-containing groups bonded to the graphene and recovered the conjugated network of the graphitic lattice.[40,41] Thus, the thermal conduction ability of graphene was recovered and the heat conduction of the composites was enhanced. Zheng et al. employed rGO to increase heat conduction in *n*-eicosane/silica ss-PCMs and the thermal conductivity increased about 193%. Conversely, the phase change enthalpy still sharply decreased owing to the interfacial interactions between PCM molecules and the solid support that restricted the normal crystallization behavior of *n*-eicosane.[42] Hence, developing high-thermal-performance ss-PCMs with less energy consumption and simple preparation process is mandatory in thermal energy sectors.

In this paper, a highly thermal conductive hierarchical 3D reduced graphene porous carbon support for PCMs was developed by carbonizing a GO@MOF-5 template, in which GO was reduced to rGO and MOF was carbonized to porous carbon simultaneously (Scheme 1). The porous carbons possess high surface area and large pore volume, which were capable of adsorbing stearic acid (SA). Furthermore, SA was stabilized in porous carbon via capillary force. The interaction between SA and rGO greatly improved the thermal conductivity owing to a significantly decreased interface thermal resistance among the constituents. Furthermore, the obtained hierarchical 3D network structure of the support was conducive to the stretching and crystallization of the SA molecule and led to enhanced heat release efficiency. The obtained ss-PCMs exhibited large phase change enthalpy, high thermal stability, significantly enhanced thermal conductivity, and transient thermal response, which showed promising potential in thermal energy storage applications.

■ EXPERIMENTAL SECTION

Materials. Dimethylformamide (AR), triethylamine (AR), H_2O_2 (AR), methanol (AR), and ethanol (AR) were purchased from Sinopharm Chemical Reagent Co., Ltd, and zinc nitrate hexahydrate (99.9%), *p*-phthalic acid (99.9%), stearic acid (Alfa), octadecane (Alfa), octadecanol (Alfa), and PEG2000 (Alfa) were purchased from J&K Scientific Ltd.

Synthesis of the Supports for PCMs. The synthesis process of MOF-5, GO@MOF-5, MOF-5-C, rGO@MOF-5-C, and rGO/MOF-5-C physical mixture is detailed in the Supporting Information. In this study, the mass ration of GO in GO@MOF-5 was constant (3 wt %).

Synthesis of Composite ss-PCMs. Composite ss-PCMs based on different supports were prepared by a wet impregnation method.[15,31,43] Typically, SA/rGO@MOF-5-C was prepared as follows: 0.9 g of SA was dissolved in 20 mL of ethanol solution and then 0.1 g of rGO@MOF-5-C was introduced into SA solution. The solution containing SA and rGO@MOF-5-C was vigorously stirred at 80 °C for 1 h, and then the mixture was dried for 12 h at 80 °C. Finally, black solid composite PCMs were obtained. Similar procedures were followed to fabricate the other composite materials by changing only the mass ratio of PCMs and the supports. Therefore, ss-PCMs with different SA percentages (40, 70, 75, 80, 85, and 90 wt %) were synthesized to study their thermophysical performance. In addition, composites with different PCMs (octadecane, PEG2000, and octadecanol) were prepared to study their general applicability. Likewise, ss-PCMs with different supports, such as MOF-5-C, MOF-5, rGO/MOF-5, and GO@MOF-5, were prepared to investigate the role of hierarchical network structures in stabilizing PCMs.

Characterizations. Scanning electron microscopy (SEM, ZEISS SUPRA55) and transmission electron microscopy (JEM-2010) were conducted to test the morphology of different materials. The crystallization of the composites was measured by powder X-ray diffraction (XRD, M21X, Cu Kα radiation at 40 kV and 150 mA, λ = 0.15406 nm). The characterization of graphene was tested by a Raman system (Renishaw inVia) with a 633 nm excitation wavelength. The thermal stability of the samples was measured with an equipment (thermal gravimetric analysis (TGA), NETZSCH, STA449F) at a 5 °C min^{-1} heating rate and under a N_2 atmosphere. Fourier-transform infrared (FT-IR) spectra were recorded on Nicolet 6700 using the KBr pellet technique. N_2 adsorption−desorption isotherms were tested by a Micromeritics ASAP 2420 adsorption analyzer. Heating and cooling performances of the PCMs were tested by differential scanning calorimetry (DSC, NETZSCH, STA449F3) with a rate of 10 °C min^{-1} under 50 mL min^{-1} N_2 flow. Heat conduction in PCMs was measured by the transient plane source method, employing a hot disk thermal constant analyzer (Hot Disk TPS 2500S). The heat storage/release curves were tested by heating the PCMs with the glass tube in a 110 °C oil bath for 900 s and then cooling down to room temperature naturally, and the data were collected by a R2100 data acquisition system.

■ RESULTS AND DISCUSSION

Characterization of Supporting Materials. GO sheets, which acted as a template for the in situ growth of MOF-5,

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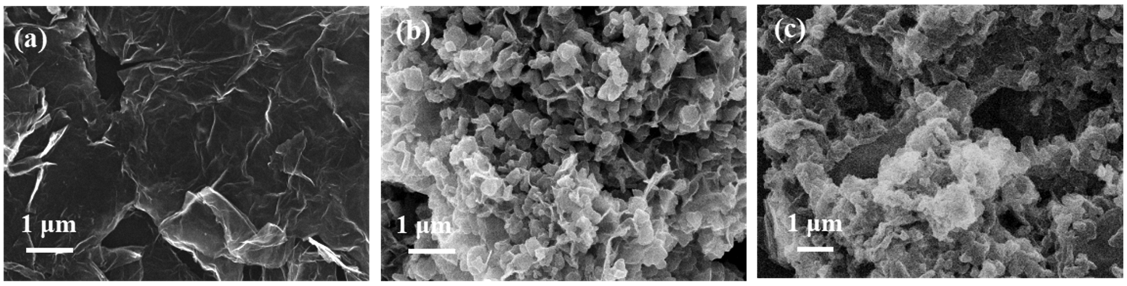

Figure 1. SEM images of (a) GO, (b) GO@MOF-5, and (c) rGO@MOF-5-C.

showed wrinkled surface textures with curling edges (Figure 1a), and GO@MOF-5 exhibited relatively homogeneous dispersion of GO in MOF-5 without any detectable agglomeration (Figure 1b). Importantly, MOF-5 stabilized the network structure of the supporting matrix. Finally, a hierarchical 3D reduced graphene/porous carbon structure was obtained by carbonizing GO@MOF-5 under a nitrogen atmosphere at 1000 °C for 3 h. The average size of MOF-5 decreased after carbonization (Figure 1c) owing to ZnO particles that were migrated, aggregated, and then vaporized under 1000 °C.

The formation of MOF-5 on GO is a key step in preparing GO@MOF-5 matrixes. All of the typical peaks of MOF-5 like those associated with symmetric and asymmetric vibrations of BDC (1610−1550 and 1420−1335 cm^{-1}) and adsorbed water (3500−3200 cm^{-1})[44] can be seen in FT-IR spectra (Figure S1a) of GO@MOF-5. Similarly, the XRD patterns (Figure S1b) of MOF-5 were consistent with those previously reported in the literature.[45] A slight left shift of the major peaks indicated that GO lengthened the bond lengths of MOF-5 without destructing its framework structures. Thus, XRD and FT-IR analyses confirmed that MOF-5 crystals were grown on the GO.

Nitrogen adsorption−desorption isotherms characterized the porous structures of the support (the data are summarized in Table 1). GO@MOF-5 presented a typical type-I adsorption

Table 1. Surface Area and Total Pore Characterstics of the as-Synthesized Samples

samples	S_{BET} (m^2 g^{-1})	V_{pore} (cm^3 g^{-1})	D_{pore} (nm)
GO@MOF-5	1034.2	0.46	0.57
rGO@MOF-5-C	2726.9	3.56	4.71

isotherm with a steep increase at lower relative pressures (Figure 2a), indicating the existence of micropores in the supports, whereas the pores of GO@MOF-5 were mainly distributed at about 0.57 nm (Figure 2b). After the carbonization process, the hierarchical 3D structural rGO@MOF-5-C was obtained, in which the GO@MOF-5 framework acted as a self-sacrificed template and facilitated the formation of the nanoporous carbon with large pore volume and high specific surface area. The specific surface area of rGO@MOF-5-C was 2726.9 m^2 g^{-1}, which was by far higher than that of GO@MOF-5 (1034.2 m^2 g^{-1}). Moreover, the total pore volume of rGO@MOF-5-C (3.56 cm^3 g^{-1}) was higher than that of GO@MOF-5 (0.46 cm^3 g^{-1}). rGO@MOF-5-C displayed the pore size distribution with a peak centering at 1.6−5 nm (Figure 2b), which was a desirable characteristic for the encapsulation of PCMs.[29,31]

GO@MOF-5 presented peaks similar to those of MOF-5 in XRD patterns. After the carbonization process, the peaks of GO@MOF-5 disappeared and rGO@MOF-5-C revealed a broadened peak located at around 26° (Figure 2c), corresponding to the (002) diffraction of the graphitic layered structure.[46] Moreover, no peak can be observed in the XRD pattern of MOF-5-C, which indicated the amorphous carbon structure.[47] The results indicated that a reduced graphene structure was successfully introduced into the rGO@MOF-5-C support.

Moreover, Raman spectra were recorded to confirm the defects of the supporting materials. Two notable peaks were observed in rGO@MOF-5-C, a typical D band at about 1319 cm^{-1} related to the defects of the graphitic lattice of the sp^3-hybridized carbon and a G band centered at 1587 cm^{-1} attributed to the presence of sp^2-bonded carbon (Figure 2d). Herein, a heightening intensity of the G peak with a graphene structure was observed, indicating a larger proportion of the ordered graphitic sp^2 hybrid carbon phase in rGO@MOF-5-C.[48,49] Moreover, the intensity ratio of the D and G bands (I_D/I_G) was established to quantify the defects in the carbons of MOF-5-C and the rGO/MOF-5-C supporting materials. The I_D/I_G ratio of MOF-5-C was 1.31, and the rGO/MOF-5-C ratio was 1.09, respectively, indicating lower density of defects in rGO/CMOF-5 than that in MOF-5-C due to the successful introduction of rGO.

Characterization of Composite ss-PCMs. Figure S2 shows the morphologies of SA/rGO@MOF-5-c ss-PCMs with different wt % of SA introduced in the rGO@MOF-5-C structure. As the wt % of PCMs increased, the structures of the ss-PCMs were more compact than those of rGO@MOF-5-C. For example, in 85 wt % SA/rGO@MOF-5-C ss-PCMs, the voids found in rGO@MOF-5-C can absorb the liquid SA. However, when the mass ratio of PCMs reached 90 wt %, the abundant pores of rGO@MOF-5-C were fully filled and the surfaces turned smooth. This indicates that the optimal mass ratio of SA in rGO@MOF-5-C was 90 wt %, in which SA was stabilized by capillary force and surface tension and did not leak above its melting point (Figure S3).

Figure S4 shows the XRD and FT-IR results of the pure PCMs, the supporting materials, and the prepared ss-PCMs. The diffraction peaks at 6.9, 20.5, 21.5, and 23.8° were attributed to SA (Figure S4a), whereas the diffraction peak at 26° was corresponding to rGO@MOF-5-C. All of the typical diffraction peaks of SA were observed in SA/rGO@MOF-5-C ss-PCMs, indicating that PCMs were successfully introduced into the solid support. Figure S4b presents the FT-IR results of SA rGO@MOF-5-C and SA/rGO@MOF-5-C ss-PCMs. All of the prominent peaks of SA including the stretching vibration of the methylene group (2919 and 2849 cm^{-1}) are present in the

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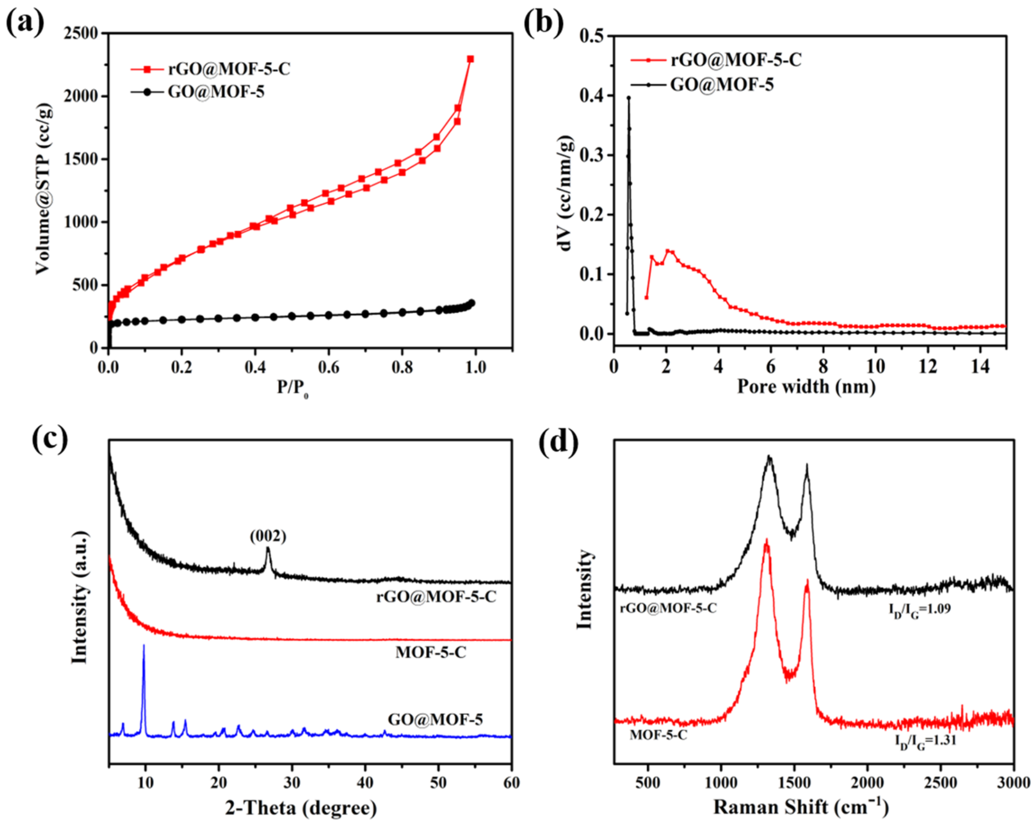

Figure 2. (a) N_2 adsorption−desorption isotherms, (b) pore size distribution, (c) XRD patterns, and (d) Raman spectra of prepared supports for PCMs.

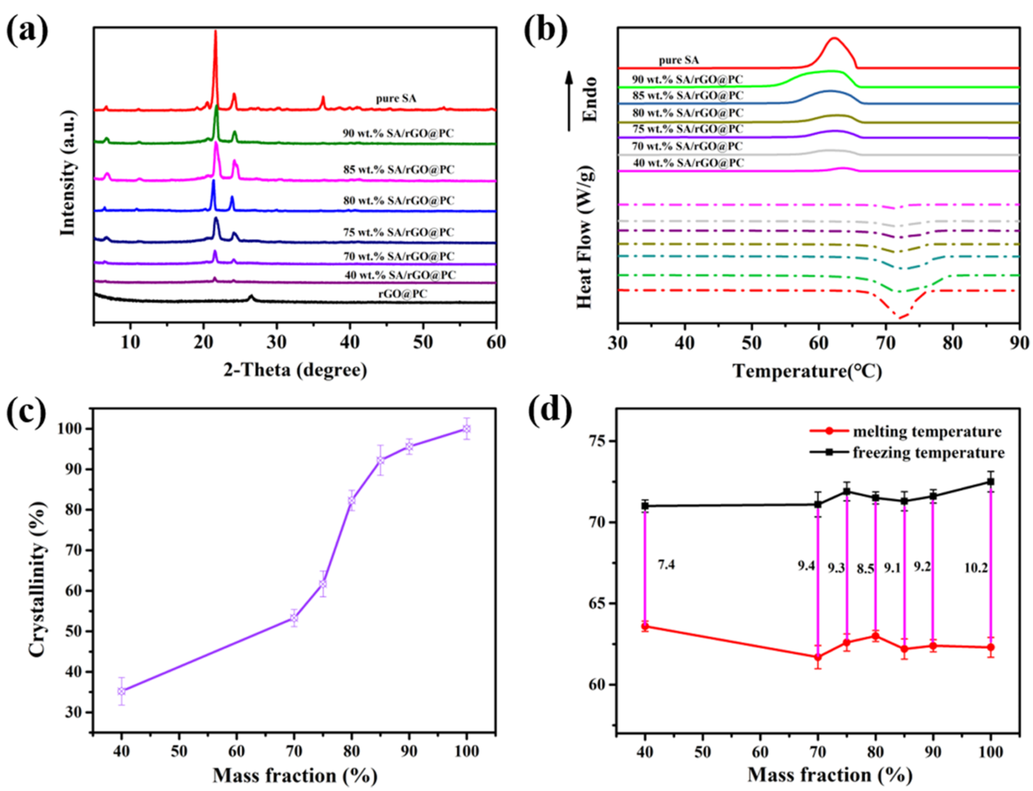

Figure 3. (a) DSC curves, (b) XRD patterns, (c) crystallinity, and (d) supercooling of the SA in ss-PCMs with various SA mass ratios.

composite PCMs, which demonstrated that SA was successfully incorporated into rGO@MOF-5-C.

Thermal Performance of ss-PCM Composites. Phase change latent heat is an essential factor in evaluating the energy

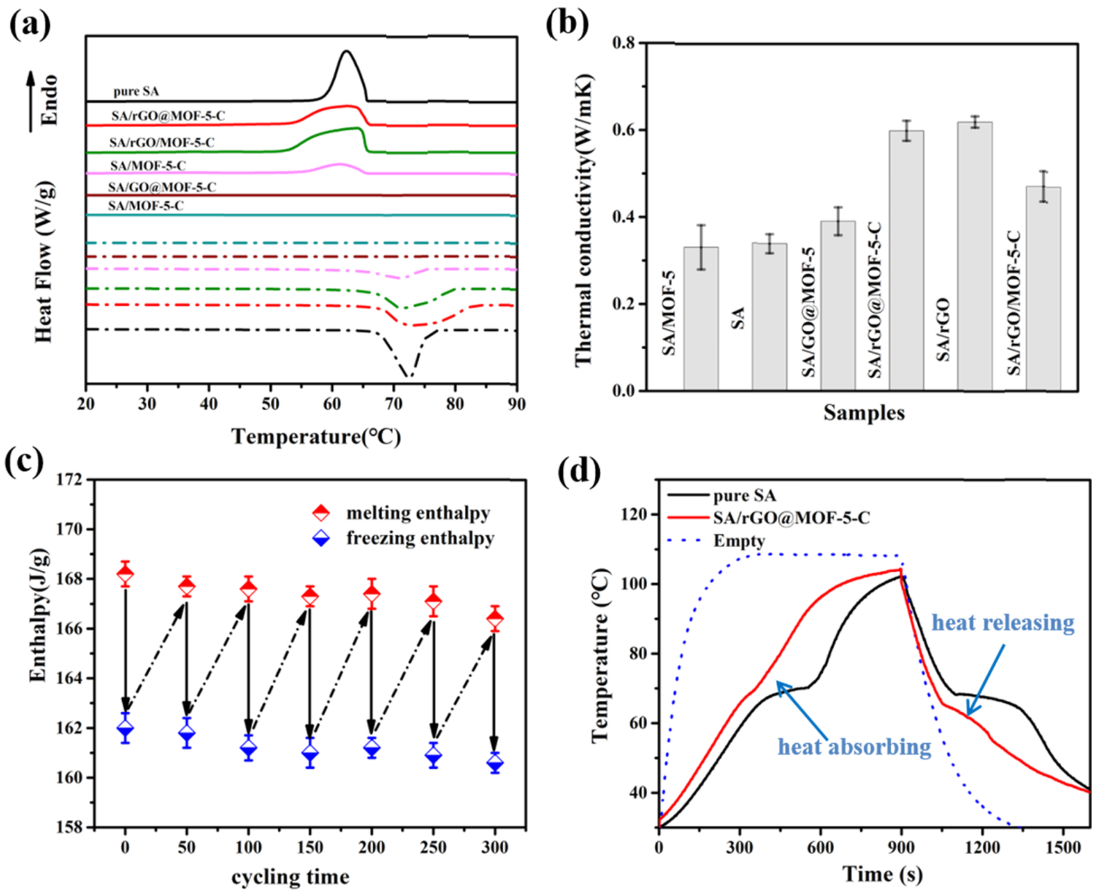

Figure 4. (a) DSC curves of ss-PCMs based on different supporting materials. (b) Thermal conductivity of the prepared ss-PCMs. (c) Melting and freezing enthalpies of SA/rGO@MOF-5-C at different thermal cycling times. (d) Temperature evolution curves of pure SA, the empty glass tubes, and SA/rGO@MOF-5-C.

Table 2. Thermal Properties of SA and Composite PCMs

samples	SA loading	T_m/T_f (°C)	$\Delta H_m/H_f$ (J g^{-1})	$\Delta H_m/H_f$ in theory (J g^{-1})
pure SA	100	72.5/62.3	196.1/187.5	
SA/rGO@MOF-5-C	90	71.6/62.4	168.7/162.2	176.5/168.8
SA/rGO/MOF-5-C	90	73.2/64.1	142.4/138.1	176.5/168.8
SA/MOF-5-C	85	70.3/61.5	87.5/81.5	166.7/159.4
SA/GO@MOF-5	65			137.3/131.3
SA/MOF-5	50			127.5/121.9

storage capacity of the composite PCMs. Endothermic and exothermic peaks of the composites get smaller with a decreased SA wt % in the ss-PCMs (Figure 3a, Table S1), which indicates that the interaction between SA and rGO@MOF-5-C presumably influences the phase change latent heat of ss-PCMs. As shown in the XRD (Figure 3b) results, the interactions between SA and supports restrict the motions of SA molecules during the phase change process and lead to a decreased crystallization of SA in the ss-PCMs. To further investigate the crystallinity of SA in the composite PCMs, the crystallization fraction (F_c) was calculated using eq 1.

$$F_c = \frac{\Delta H_m}{\Delta H_{pure} \Delta W} \times 100\% \tag{1}$$

where ΔH_m is the actual phase change enthalpy of the ss-PCMs, ΔH_{pure} is the enthalpy of the pure SA, and W represents the mass ratio of SA in ss-PCMs. The parameter F_c served as an indication of the interaction between the PCMs and the supporting materials. A higher F_c value demonstrates higher preservation of the crystalline phase and consequently weaker interaction. When the weight percentage of SA was increased, the F_c increased correspondingly (Figure 3c), indicating that a smaller fraction of SA was restricted by the support. When rGO@MOF-5-C stabilized 90 wt % SA, the F_c of ss-PCMs reached up to 95.6 ± 2.6%, suggesting that most of the SA segments can move freely.

The supercooling phenomenon demonstrated the thermal transport characteristics of the ss-PCMs (Figure 3d). The weak interactions including hydrogen bonding and van der Waals forces between SA and rGO@MOF-5-C slightly reduced the melting point of SA.[38] The supercooling extents of the prepared ss-PCM with 40, 70, 75, 80, 85, and 90 wt % SA were 27.5, 7.8, 8.8, 16.7, 10.8, and 9.8% lower than those of pristine SA, respectively. This implied that rGO@MOF-5-C favorably reduced the supercooling extent of SA, which is imperative for the practical application. The hierarchical 3D reduced

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graphene/porous carbon structure provided thermal conduction paths for ss-PCMs and consequently accelerated the rate of phase change. This indicates that the heat from the environment can be transferred to the inner side of the ss-PCMs more easily due to the enhanced thermal conductivity of rGO@MOF-5-C, resulting in the fast phase transition process of ss-PCMs.[17]

To demonstrate that the hierarchical 3D network carbon structure is in favor of stabilizing SA, four kinds of supporting materials (MOF-5, GO@MOF-5, MOF-5-C, and rGO/MOF-5-C) were also investigated (Figure 4a). Each supporting material revealed different PCM adsorption ability and the crystallization fraction of the SA in the composites (the data are summarized in Table 2). MOF-5 has a small loading amount of SA (50 wt %) most probably owing to the small pore size that is used for PCM phase transition.[31] SA was adsorbed and restricted on the surface of microporous MOFs; thus, there was no phase change latent heat of SA/MOF-5 ss-PCMs. When GO was introduced in MOF-5, the loading content of SA in GO@MOF-5 was increased to 65 wt % due to the formation of a network structure. However, the micropores of GO@MOF-5 still limit both the loading content (65 wt %) and phase change enthalpy (0 J g^{-1}) of ss-PCMs. After the carbonization process, MOF-5 was transferred to a hierarchical porous carbon with micropore and mesopore characteristics (Figure S5); thus, the loading amount of SA in MOF-5-C reached 85 wt %. Nevertheless, the crystallization fraction of the SA in MOF-5-C was only 56.9 ± 1.9% because a large number of SA molecules were still restricted in the micropores of the supporting material. When rGO was homogeneously dispersed in MOF-5-C, it alleviated the interfacial effect between SA and the solid support and encapsulated high content of PCMs (90 wt %) with the crystallization fraction reaching up to 95.6 ± 2.6%. These demonstrated that the 3D structure is conducive to the stretching and crystallization of the SA molecule and allows a relatively free movement in the matrix. However, the phase change enthalpy was lower than its theoretical value and the crystallization fraction of the SA was 80.7 ± 2.5%. This demonstrates that the confined pore space of physically mixed rGO and MOF-5-C reduces the mobility of SA molecules and inhibits the phase change of SA. To determine the general applicability of the prepared support, rGO@MOF-5-C was applied to stabilize different PCMs, such as octadecane, PEG2000, and octadecanol (Figure S6). All of the obtained ss-PCMs showed large phase change enthalpy (Table S2), further demonstrating the extensive suitability of the prepared supporting material in preparing ss-PCMs.

Thermal gravimetric analysis (TGA) was conducted to investigate the thermal stability of the PCMs. SA started to decompose at 180 °C and terminated at 380 °C (Figure S7). rGO@MOF-5-C exhibited excellent thermal stability under 800 °C, which proves its competency in stabilizing SA. As expected, SA/rGO@MOF-5-C exhibited a combined thermal stability of SA and rGO@MOF-5-C. Therefore, the SA/rGO@MOF-5-C ss-PCMs were thermally stable below 180 °C. In addition, the actual impregnation ratio of SA relying on the weight loss during the decomposition is 90.8%, which is basically consistent with the theoretical loading value. The results confirmed the homogeneous introduction of SA into the supporting material without chemical reaction during the blending and impregnation process.[28]

Thermal conductivity is an important parameter to evaluate the heat transfer capability of composite PCMs. The thermal conductivity of SA/MOF-5 was 0.33 ± 0.05 W m^{-1} K^{-1}, which was smaller than that of pure SA (0.34 ± 0.02 W m^{-1} K^{-1}) (Figure 4b). This is because most of SA molecules were absorbed on the MOF surface, and the air in MOF-5 channels blocked the thermal transformation. When a graphene structure was introduced in the support, SA/GO@MOF-5 exhibited an enhanced heat conduction of 0.39 ± 0.03 W m^{-1} K^{-1}. However, disordered scattering of phonon transport from oxygen-containing groups bonded to the graphene sheets still limits the thermal transport efficiency.[50] After the heat treatment procedure, GO was reduced to high-thermal-performance rGO and MOF-5 was simultaneously carbonized to porous carbon, and the heat conduction of SA/rGO@MOF-5-C was improved to 0.60 ± 0.02 m^{-1} K^{-1}. The interaction between SA and rGO significantly decreased the interfacial thermal resistance, whereas enhanced the heat conduction in composite PCMs (thermal conductivity of SA/rGO was 0.62 ± 0.01 W m^{-1} K^{-1}). Furthermore, the 3D structure of rGO@MOF-5-C expanded phonon transformation from nanoscale to macroscale by facilitating the heat transformation between the composite PCMs and the outside environment. The composite PCMsSA/rGO/MOF-5-C (90 wt %) exhibited a decreased thermal conductivity (0.47 ± 0.04 W m^{-1} K^{-1}), most probably owing to the encapsulation characteristics that part of PCMs was absorbed in an isolated MOF-5-C, implies far from rGO heat transfer media. Hence, the heat conversion between SA and rGO was relatively lowered. Isolated MOF-5-C blocked thermal conduction pathways of rGO and increased thermal resistance between SA and rGO/MOF-5-C. This indicates that the hierarchical 3D reduced graphene porous carbon structure of rGO@MOF-5-C significantly improved the thermal conductivity of ss-PCMs.

A heat storage/release test was conducted to investigate the potential application of the obtained composite PCMs (Figure 4d). First, the composites were loaded in the glass tubes and a temperature probe was set in the center of the samples; then, the glass tubes were heated for 900 s at 110 °C. Finally, the glass tubes were removed from the heating resources and cooled down to room temperature. The temperature evolution of the composites was collected by a data acquisition system. It took 280 s for the empty glass tubes to increase the temperature from 35 to 108 °C and 300 s for the cooling process from 108 to 35 °C. No phase change phenomenon was observed on the empty glass tubes; thus, the influence of glass tubes can be neglected in the heat storage/release tests. In addition, it took about 320 s for pure SA heating up from 35 °C to the melting point (~70 °C). The phase transition of SA from solid to liquid lasted for about 150 s. Likewise, the temperature increased from 70 to 105 °C in about 280 s. When the glass tubes were removed from the heating source, the temperature of SA decreased to the freezing temperature of 68 °C within 190 s and SA changed from liquid to solid in ~200 s. Consequently, the temperature decreased to room temperature within 150 s. On the other hand, SA/rGO@MOF-5-C took 280 s for increasing the temperature from 35 to 70 °C and 50 s for adsorbing the heat, which is shorter than that for pure SA persumably because of the rapid heat transfer rate capability of the composite PCMs. Similarly, during the heat releasing process, compared with that for pure SA, the time taken by the composites to cool down to the phase change temperature was decreased to ~150 s and the phase change time was reduced to ~160 s. Therefore, the obtained SA/rGO@MOF-5-C

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composites show a transient thermal response that is suitable for thermal energy storage applications.

Thermal cycling performance is an essential evaluation indicator in the development and application of composite PCMs. SA/rGO@MOF-5-C has been cycled 300 times. After 300 cycles of melting and freezing processes, no significant changes were observed in DSC curves (Figure S8) and the corresponding phase change enthalpy (Figure 4c), suggesting the superior stability of prepared SA/rGO@MOF-5-C composites. In addition, the thermal conductivity of the composites after cycling 300 times was 0.61 ± 0.02 m^{-1} K^{-1}, which was consistent with the result before recycling. This indicates that the porous support has excellent structural stability and the 3D continuous heat transfer paths are not disrupted during the repeated heat charge/discharge process. The SEM images (Figure S9) of composite PCMs before and after the recycling process had no obvious differences; the main characteristic peaks of XRD (Figure S10a) and FT-IR (Figure S10b) spectra of the composites and the temperature evolution curves (Figure S11) in the heat storage/release test show almost the same results. All of the aforementioned results indicate that rGO@MOF-5-C is a favorable porous support for SA stabilization and the obtained composite PCMs have a great thermal recycling performance.

CONCLUSIONS

In summary, a rGO@MOF-5-C support for PCMs has been successfully synthesized by carbonizing a GO@MOF-5 template. During the carbonization process, GO was reduced to rGO, whereas MOF-5 was carbonized to porous carbon and resulted in a hierarchical 3D reduced graphene/porous carbon structure. The highly porous carbons with large specific surface area were favorable for adsorbing PCMs. Moreover, PCMs were stabilized in the pores of carbon through capillary forces and interactions between SA and rGO@MOF-5-C with a significantly decreased interface thermal resistance, thereby improving thermal conductivity. Furthermore, the hierarchical 3D network structure of the supporting material improved the crystallization behavior of the SA with enhanced heat release efficiency. Compared with the rGO/MOF-5-C physical mixture, the hierarchical 3D structure of rGO@MOF-5-C improved the thermal conductivity up to 27.7% (0.60 ± 0.02 W m^{-1} K^{-1}) with increased phase change latent heat of 168.7 J g^{-1} (18.5%). In addition, the obtained ss-PCMs exhibited transient thermal response and good durability, indicating their potential applications in thermal energy storage.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.8b09541.

Experimental section of all samples; FT-IR, XRD, and the N_2 adsorption−desorption isotherms of the supports; digital photographs, SEM, and the thermal properties of the different composite PCMs; FT-IR, XRD, DSC, and temperature evolution curves of the composite PCMs before and after recycling (PDF)

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Notes

The authors declare no competing financial interest.

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